

SQL Music Store Analysis

Business Insights using PostgreSQL

GITHUB PORTFOLIO

DATA ANALYST PROJECT



Project Overview

This project analyzes the **Chinook Music Store Database** using PostgreSQL to answer real business questions through structured SQL queries — from customer behavior to revenue trends.

Analysis Objectives

- Customer Analysis
- Revenue Analysis
- Artist & Genre Performance
- Country-wise Sales

Technology Stack



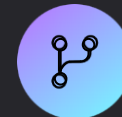
PostgreSQL



SQL

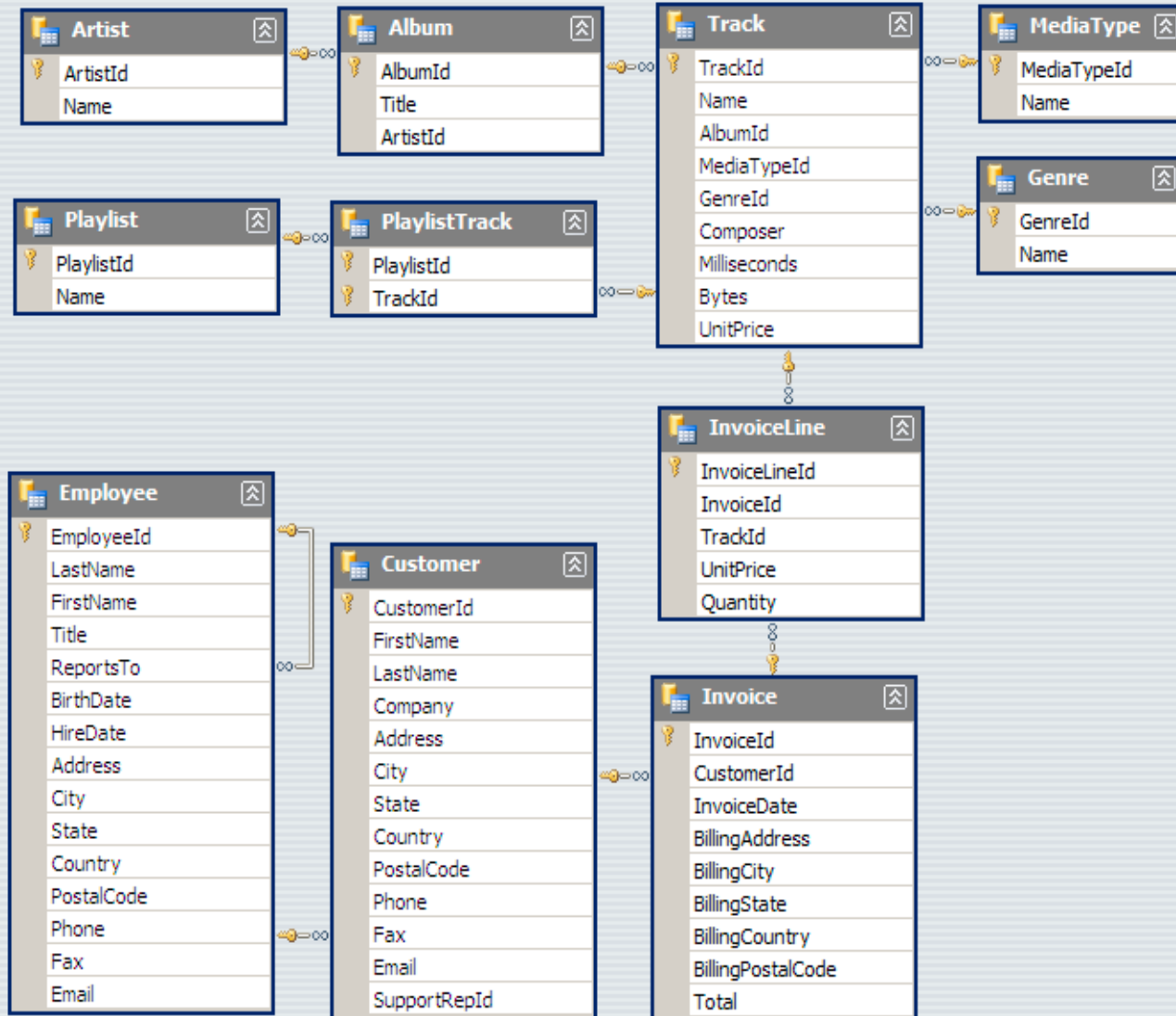


pgAdmin



Git & GitHub

Database Schema



Core Tables

Employee & Customer

People in the system

Invoice & InvoiceLine

Purchase transactions

Track, Album & Artist

Music catalog

Genre

Music classification

- The database follows a relational structure where customers purchase tracks through invoices, enabling analysis of sales, artists, albums, genres, and customer behavior.

PostgreSQL Features Used

These PostgreSQL features were applied to solve practical business analysis problems across the dataset.

SELECT & WHERE

Filtering and retrieving data

ORDER BY & LIMIT

Sorting and limiting results

GROUP BY

Aggregating by category

COUNT, SUM, AVG

Aggregate calculations

INNER JOIN

Multi-table relationships

Subqueries

Nested query logic

CTE (WITH)

Readable query structure

Window Functions

Advanced analytics

ROW_NUMBER()

Ranking and partitioning

Easy SQL Queries

Foundational queries that establish baseline business insights from the Chinook database.

1

Senior-most Employee

Identify the employee with the longest tenure using ORDER BY on hire date.

2

Countries with Most Invoices

Group invoices by billing country to find top markets.

3

Top 3 Invoice Totals

Rank invoices by total amount to identify highest-value transactions.

4

Best Revenue City

Aggregate invoice totals by city to find the top-performing location.

5

Best Customer

Determine the customer with the highest cumulative spend.

Moderate SQL Queries

Intermediate queries combining filters, aggregations, and joins for deeper insights.



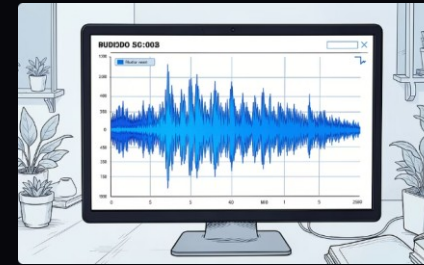
Rock Music Listeners

Filter tracks by genre and count customers who purchased rock music.



Top Rock Artists

Join artist, album, and track tables to rank rock artists by sales volume.



Tracks Longer Than Average

Use a subquery to calculate average track length and filter above it.

Advanced PostgreSQL Queries

Complex queries using CTEs, window functions, and multi-table JOINS to solve sophisticated business questions.

Customer Spending by Artist

Rank customers by total spend per artist using window functions.

Most Popular Genre by Country

Use CTEs and ROW_NUMBER() to find the top genre per country.

Top Customer by Country

Partition customers by country and rank by invoice total.

Query Flow



This pipeline transforms raw relational data into ranked, actionable business insights.

Key Business Insights

Five data-driven findings from the analysis with direct strategic implications.



USA Dominates Sales

The USA leads in both invoice count and revenue, representing the primary market for expansion.



Rock Is #1 Genre

Rock generates the highest revenue and has the most listeners across all countries.



Top Revenue City

One city stands out as the highest revenue generator, ideal for targeted marketing.



Best Customer Identified

The highest-value customer was identified for VIP retention strategies.



Genre Preferences Vary by Country

Genre popularity differs significantly across countries, enabling localized campaigns.

Challenges & Solutions

Real-world SQL analysis comes with complexity. Here's how each challenge was addressed.



Complex Joins

Used table aliases and CTEs to simplify multi-table relationships and improve readability.



Aggregation Logic

Applied GROUP BY with HAVING clauses to filter aggregated results accurately.



Subquery Optimization

Replaced nested subqueries with CTEs for cleaner, more maintainable query structure.



Ranking & Partitioning

Leveraged ROW_NUMBER() and window functions to rank customers and genres without self-joins.



Project Summary & Next Steps

What Was Accomplished

- Analyzed the Chinook Database with PostgreSQL
- Wrote 11 SQL queries across three difficulty levels
- Used CTEs, window functions, and multi-table JOINS
- Delivered business insights on revenue, customers, and genres

What's Next

- Build interactive dashboards in Power BI or Tableau
 - Perform cohort and trend analysis on the data
 - Automate reporting with Python and SQLAlchemy
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